James Foehner

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Portfolio | GitHub | LinkedIn | Live filing monitor

EDUCATION

Northeastern University | Khoury College of Computer Sciences

BS in Data Science, expected May 2027 | Boston, MA

Coursework: Machine Learning & Data Mining, Algorithms, Programming in C++, Statistics for Economists

SKILLS

Financial analysis: Excel financial modeling, comparable-company and sum-of-the-parts valuation, event studies

Data and programming: Python, SQL, R, C++, WRDS (CRSP, Compustat), SEC EDGAR, MySQL

Methods: Predictive modeling, walk-forward validation, LLM extraction and evaluation, bootstrap testing

SELECTED PROJECTS

Activist Target Screener and Post Holdings Investment Pitch | Oct 2026

Python, *SQL*, *WRDS (CRSP, Compustat)*, *SEC EDGAR*, *Excel* | Code

- Collected 232 activist campaigns from SEC 13D filings; a 227-campaign event study found 7.0% abnormal announcement returns (t = 13.3), with one-year gains concentrated in targets later acquired (+19% vs. ~0%).
- Trained a gradient-boosting screener on 44,000 firm-years using walk-forward validation; top-decile predictions captured 27% of future targets (AUC 0.73). Ranked E2open in the top 0.4% before Elliott's 2023 campaign.
- Identified Post Holdings, ranked 7th among 2,600+ companies, and modeled a Weetabix sale and debt reduction against debt-funded buybacks; estimated ~$111/share versus $77, with stronger downside performance.
- Tested deleveraging re-rating and spin-off assumptions using data on 182 spin-offs since 2010; found neither held and revised the investment memo around capital allocation.

Activist Filing Monitor | Nov 2026

Python, *Streamlit*, *Claude API*, *SEC EDGAR* | Live app

- Deployed a tool that tracks activist SEC filings and classifies investor demands; linked results to target scores, campaign histories, and recent filings for 2,600 companies.
- Evaluated extraction against 89 hand-labeled filings (0.84 macro-F1); built 26 unit tests and displayed source text alongside labels for review.

LLM Fine Tuning and Reward Evaluation | Nov 2026

PyTorch, *Hugging Face*, *LoRA*, *GRPO* | Code

- Fine-tuned a small language model for filing analysis, raising macro-F1 from 0.35 to 0.67; tested reward failures and used paired bootstrap analysis to assess incremental gains.

NBA Prediction AI | Jan–Apr 2026

Python, *scikit-learn*, *Streamlit*, *Plotly*

- Built a pipeline on ~80,000 play-by-play rows and 30 features; gradient boosting achieved 0.84 ROC-AUC and 0.071 Brier score, improving AUC by 0.08 over logistic regression.
- Deployed an interactive Streamlit app for model inference and sensitivity analysis.

Aircraft Incident Database and Analytics | May–Jul 2025

R, *MySQL*, *SQL*

- Designed a 3NF database and R ETL pipeline using batched inserts and transactions; built SQL views and stored procedures and validated counts and aggregates against source data.